ELECTRICAL—LOW TIRE WARNING INDICATOR ILLUMINATED—NO DIAGNOSTIC TROUBLE CODES (DTC) PRESENT	Article No. 02-7-5

FORD: 2001-2002 WINDSTAR

ISSUE

Some vehicles built before 3/3/02 may exhibit a Low Tire Warning lamp on with no Diagnostic Trouble Codes (DTC) present, incorrectly indicating a low tire situation with no low tire. This may be caused by current internal programming of the Anti-Lock Brake Control (ABS) Module sensing very slight differences in tire speed inputs allowing the warning lamp to illuminate.

ACTION

Verify condition. If normal diagnostics cannot pinpoint the condition and no DTC's are present, refer to the following Service Information to help diagnose and correct the condition.

SERVICE INFORMATION

System Function:

The low tire warning system utilizes very small differences in ABS wheel speed information to estimate whether or not a single tire has lost between 10% and 30% of the tire pressure. The system's logic requires that the revolutions per mile of each of the four tires change in the same manner as the vehicle speed changes throughout the driving range, and as vehicle loading conditions are changed.

Customers will often experience incorrect indications of low tire pressure after a single tire has been damaged and replaced. This is more prevalent with vehicles equipped with the Uniroyal Tiger Paw tire if customers replace the Tiger Paw with a P-metric size designation. Note that the OEM Uniroyal Tiger Paw tires are 215/65-R16, and are not P-metric (P215/65-R16). Also note that the P-metric and non-P-metric type tires have different load ratings. A similar situation can also occur when the customer replaces a single tire with an entirely different tire.

The above single tire replacement situation can be corrected by replacing the Anti-Lock Brake Control Module portion of the ABS unit where the computer program resides, or by purchasing an exact replacement tire.

Replacing the ABS Module will not correct the following conditions labeled 1 through 6.

1. Customer did not press and hold the reset after checking/adjusting tire pressure or after service.
2. Customer had an oil change in a quick-lube facility, and tire pressure was adjusted as part of the multiple point service check, and customer did not press and hold the reset button.
3. Tire pressure reset button was not pressed at the time of vehicle delivery.
4. Tires were rotated without pressing and holding the tire reset button.
5. Tire reset was pressed with tire pressures not set to recommended pressure found on the Certification Label.
6. Slow leaks - for example, the light may illuminate once a week, but the leak may be difficult to find.

When diagnosing the Low Tire Warning system, please note the following:

1. After pressing reset, the system must learn a parameter that describes how the four tires on the system behave together. This process can require up to 32 kilometers (20 miles) depending on driving conditions. Learning takes place in a shorter amount of time by driving on straight roads at constant speeds, with minimal accelerating and braking.
2. The system requires the vehicle to be driven to detect a loss of tire pressure. Smaller amounts of pressure loss in the order of 5 psi may take longer to detect than larger pressure losses in the order of 10-15 psi.
3. The system may not function properly under the following conditions:
   a. Uneven tread wear.
   b. Driving on loose or low traction surfaces such as gravel, snow, or slush.

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c. Using tire chains.

d. Initial tire pressures are out of the specified range.

e. Two or more under-inflated tires.

f. Sudden loss of tire pressure.

g. Vehicle speeds less than 20 mph (30 km/h), greater than 120 km/h (70 mph) or driving duration less than 16 kilometers (10 miles).

h. Transporting a heavy load or towing a trailer.

4. The Low Tire Warning system may be turned off.

ABS Module Replacement

If normal diagnostics cannot pinpoint the condition and no DTC's are present and none of the previous conditions apply, replace the Anti-Lock Brake Control Module Part with part number 2F2Z-2B373-BA for non traction control vehicles and 2F2Z-2B373-CA for vehicles with traction control per Section 206-09A of the Workshop Manual.

PART NUMBER	PART NAME
2F2Z-2B373-BA	ABS Module - Non Traction Control
2F2Z-2B373-CA	ABS Module - With Traction Control

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 206000, 306000